

Evaluation of MLIPs on energy and forces

Heated DeePMD trajectories test

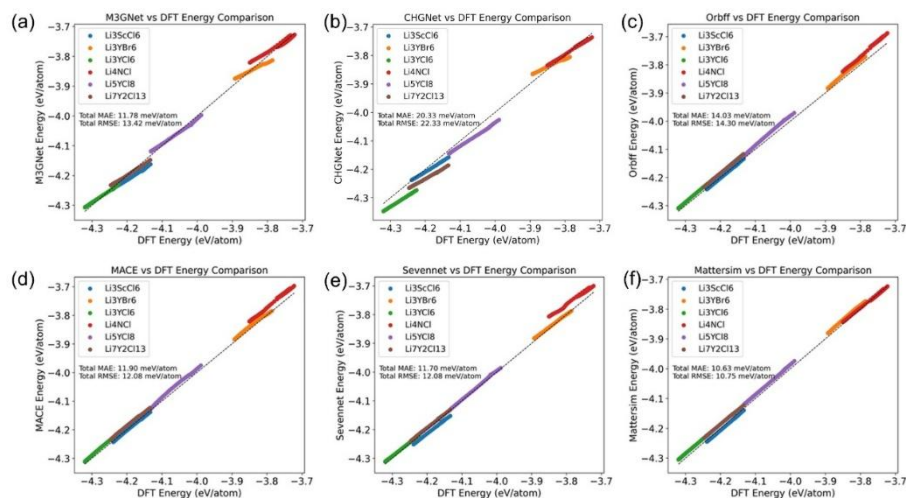


Figure S1 evaluates the performance of six MLIPs in calculating the energy of halide electrolytes at different structures sampled from heated DeePMD trajectories ranging from 100 K to 1100 K, compared to DFT.

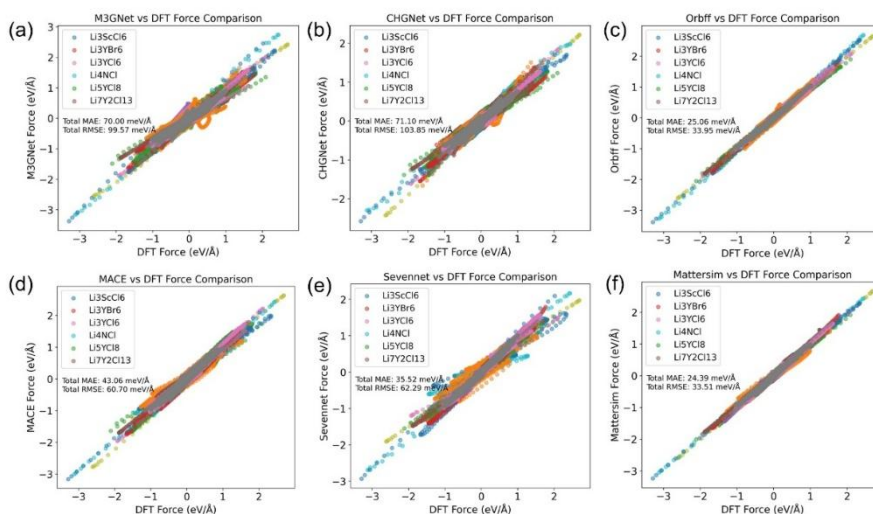


Figure S2 compares the force prediction errors of six MLIPs and DFT for various halide electrolyte structures from DeePMD trajectories (heated from 100K to 1100K).

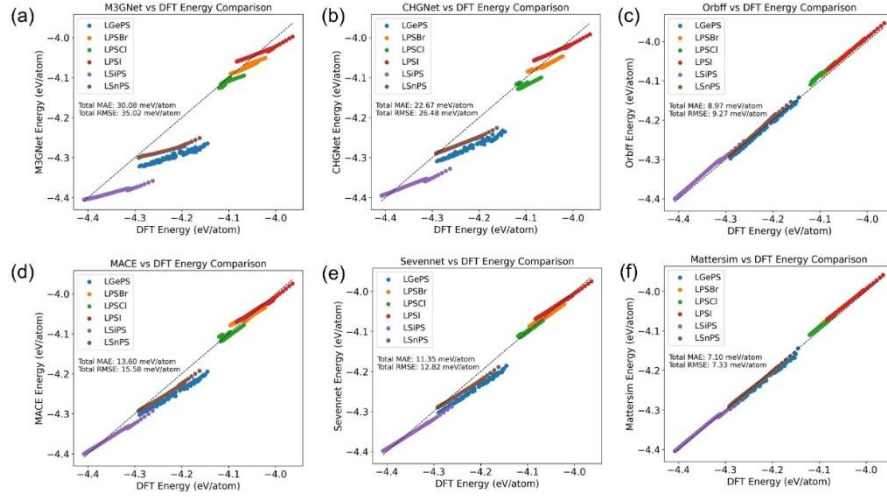


Figure S3 evaluates the performance of six MLIPs in calculating the energies of different sulfide SSE structures along heated DeePMD trajectories from 100 K to 1100 K, compared with DFT calculations.

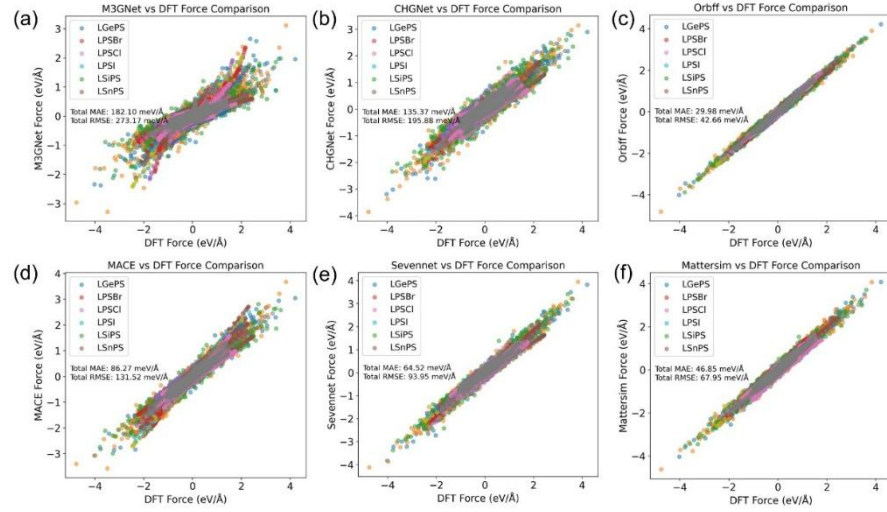


Figure S4 evaluates the performance of six MLIPs in calculating the forces of different sulfide SSE structures along heated DeePMD trajectories from 100 K to 1100 K, compared with DFT calculations.

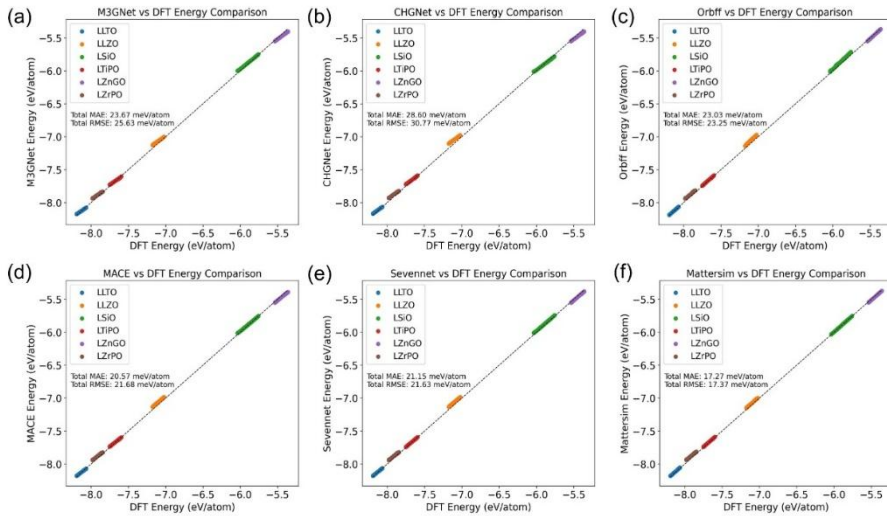


Figure S5 evaluates the performance of six MLIPs in calculating the energies of different oxide SSE structures along heated DeePMD trajectories from 100 K to 1100 K, compared with DFT calculations.

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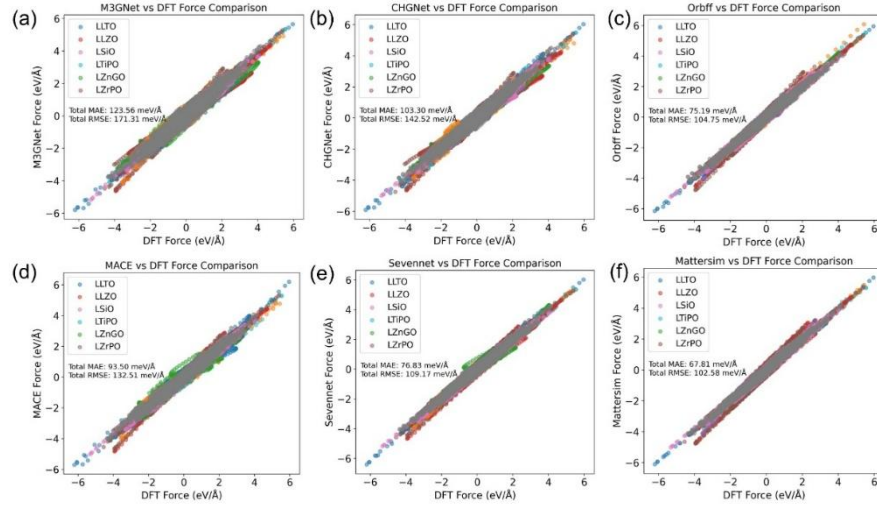


Figure S6 (a)-(f) evaluates the performance of six MLIPs in calculating the forces of different oxide SSE structures along heated DeePMD trajectories from 100 K to 1100 K, compared with DFT calculations.

Near-equilibrium trajectories test

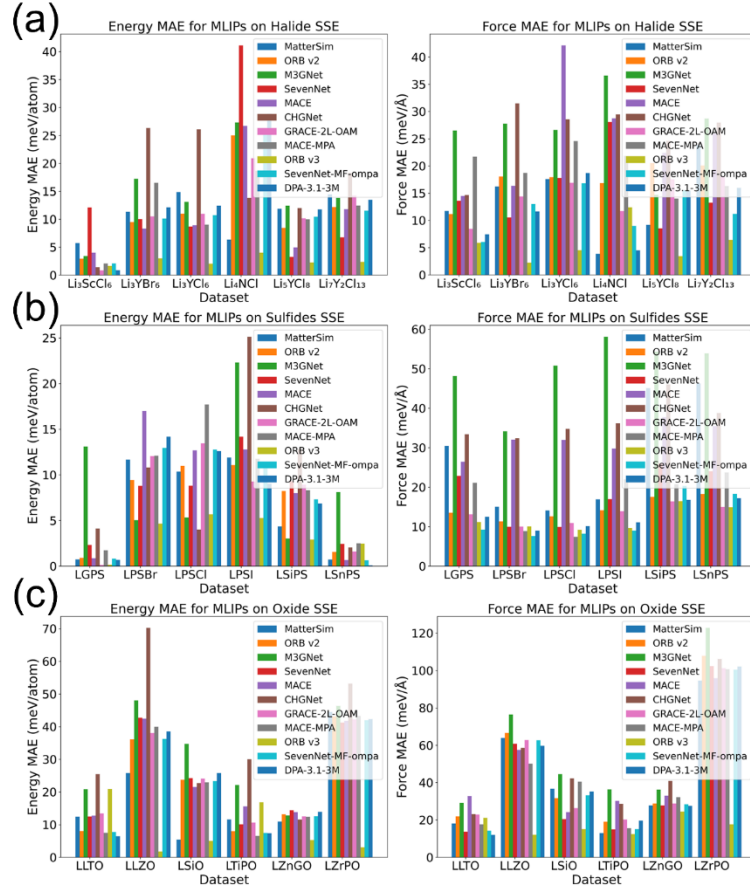


Figure S7 (a)-(c) compares the performance of 11 MLIPs with DFT in predicting energies and forces for halide, sulfide, and oxide SSE structures during 10-step DeePMD simulations at 50 K (near-equilibrium configurations).

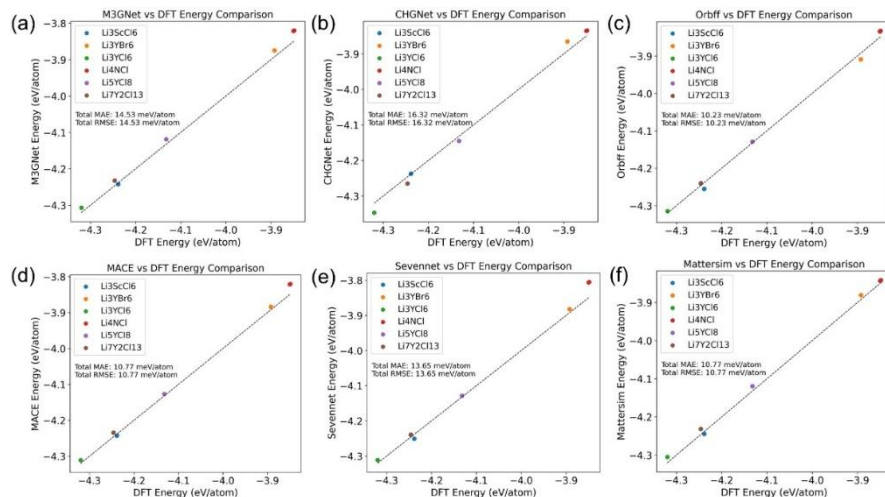


Figure S8 compares DFT-calculated energies with predictions from six MLIPs for the halide electrolyte dataset. The dataset comprises representative halide electrolytes and was generated through 10-step DeePMD simulations at 50 K to ensure near-equilibrium configurations.

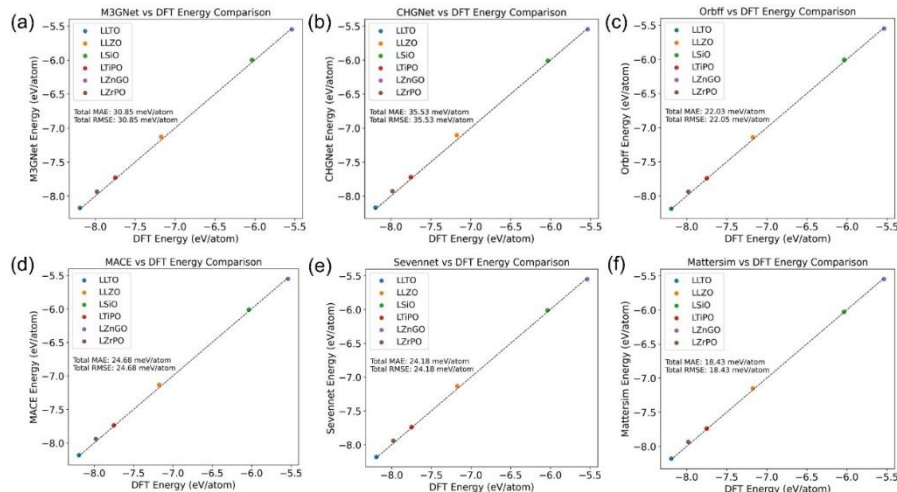


Figure S9 compares the energy predictions of six MLIPs on the oxide dataset with DFT calculations. The dataset includes typical oxide electrolytes and was generated by running 10 steps of DeePMD simulation at 50 K to ensure near-equilibrium configurations.

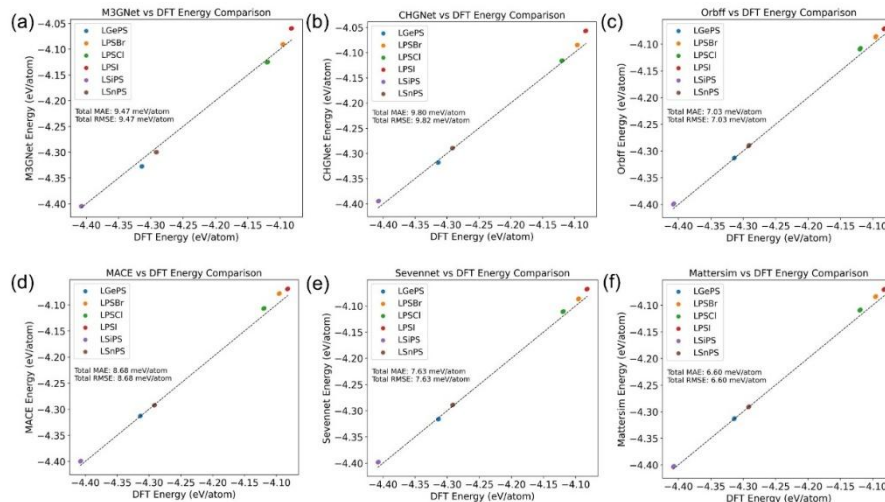


Figure S10 compares the energy predictions of six MLIPs on the sulfide test dataset with DFT calculations. The test dataset includes typical sulfide electrolytes and was generated by running 10 steps of DeePMD simulation at 50 K to ensure near-equilibrium configurations.

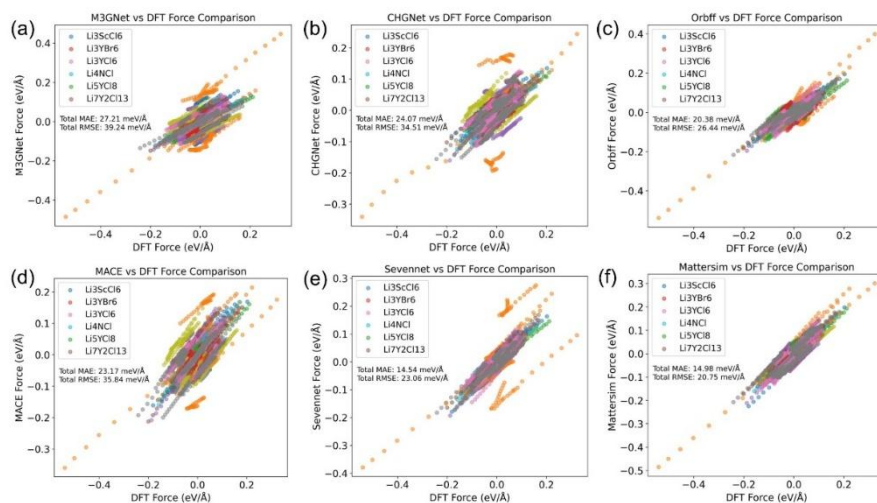


Figure S11 compares the force predictions of six MLIPs on the halide test dataset with DFT calculations. The test dataset includes typical halide electrolytes and was generated by running 10 steps of DeePMD simulation at 50 K to ensure near-equilibrium configurations.

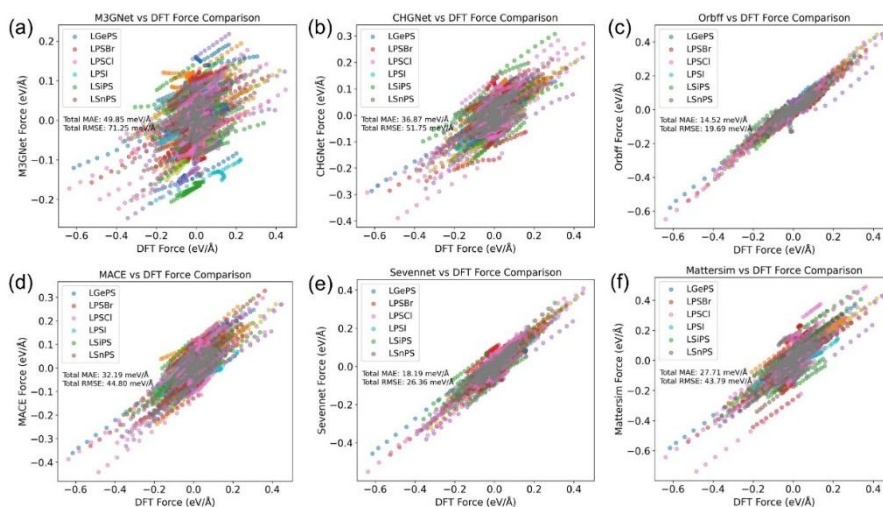


Figure S12 compares the force predictions of six MLIPs on a sulfide test dataset with DFT calculations. The test dataset includes typical sulfide electrolytes and was generated by running 10 steps of DeePMD simulation at 50 K to ensure near-equilibrium configurations.

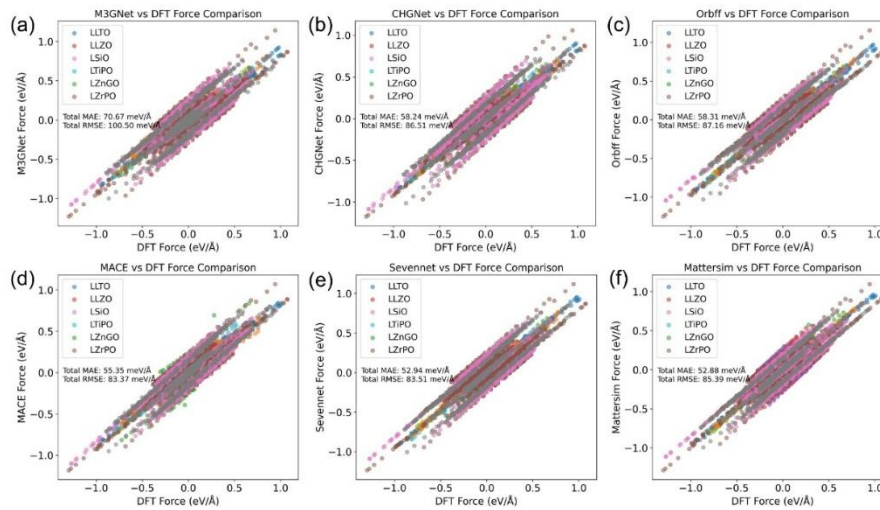


Figure S13 compares the force predictions of six MLIPs on the oxide test dataset with DFT calculations. The dataset includes typical oxide electrolytes and was generated by running 10 steps of DeePMD simulation at 50 K to ensure near-equilibrium configurations.